MIL-T-62046C(AT)
18 August 1982
SUPERSEDING
MIL-T-62046B(AT)
12 March 1969

MILITARY SPECIFICATION

TRANSMISSION ASSEMBLY: FOR TRUCK, CARGO, 6X6 M561 W/E AND TRUCK, AMBULANCE, 1 1/4 TON, 6X6, M792 W/E

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the transmission assembly to be used in Truck Cargo M561 and Truck Ambulance M792.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

P-C-437	- Cleaning Compound, High Pressure (Steam Cleaner).
TT-C-490	- Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
TT-E-529	- Enamel, Alkyd, Semi-Gloss.
TT-P-636	- Primer Coating, Alkyd, Wood, and Ferrous Metal.

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-P-514	- Plates: Identification, Instruction and Marking, Blank.
MIL-L-2105	- Lubricating Oil, Gear, Multi-purpose.
MIL-T-62047	- Truck, Cargo, 1 1/4 Ton, 6x6, M561, and Truck Ambulance, 1 1/4 Ton, 6x6, M792.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-193	- Paint Procedures and Marking for Vehicles, Construction Equipment, and Material Handling Equipment.
MIL-STD-810	- Environmental Test Methods for Aerospace and Ground Equipment.

2.1.2 Other Government documents, Government drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ARMY

11601560	- Transmission.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity, or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article (initial production) transmission. The transmission selected as first production shall be fully representative of transmissions proposed to be furnished under the contract, with all current modifications included in, and on the selected transmission, and produced on production tooling. If submitted transmission meets all requirements of this specification, no modifications shall be applied to subsequent transmissions produced under contract without prior Government approval (see 6.3).

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3.1.1 Reliability. The workmanship shall be of such quality to assure transmission assembly and components shall be free of defects that limit or reduce the transmission assembly capabilities herein expressed. Unless otherwise specified (see 6.2), transmissions made in accordance with this specification shall be subjected to examinations and tests to prove transmission capabilities to perform in accordance with the following: the transmission shall require only organizational maintenance and shall not require replacement of components or major overhaul during 20,000 miles of vehicle operation.

3.2 Materials. Materials shall be as specified in drawing 11601560. Material shall be free of defects which adversely affect performance or serviceability of the finished product (see 6.6).

3.2.1 Lubricant. Lubricant used in the transmission assembly shall conform to MIL-L-2105.

3.3 Construction. The transmission shall conform to drawing 11601560. When installed in an M561 truck or an M792 ambulance, it shall provide before the transfer case a selection of four forward gear ratios and one reverse gear ratio (see table I), sufficient to meet the performance requirements of the M561 truck and M792 ambulance. All parts requiring identification shall be identified in accordance with MIL-STD-130.

TABLE I. Transmission ratios.

Gear	Ratio
First	7.059 : 1
Second	3.581 : 1
Third	1.711 : 1
Fourth	1.000 : 1
Reverse	6.784 : 1

3.3.1 Interchangeability of parts. Component parts and assemblies of each transmission shall be constructed so that any part(s), except those assemblies requiring individual select fitting of components, may be installed in any like transmission without modification.

3.3.2 Gears and splines. All gears and working splines shall meet all performance requirements of transmission.

3.3.3 Seals and vents. The input (clutch) shaft, output shaft, shift lever shafts, and all other joints and openings shall be sealed. Seals shall prevent water and contaminant entry as well as lubricant leakage. Provision shall be made for the attachment of a vent pipe.

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3.3.4 Power take-off. Provisions shall be made for the installation of a power take-off unit.

3.4. Break-in run. Each transmission produced shall be subjected to a break-in run (see 4.4).

3.5 Performance. The completely assembled transmission shall meet all performance requirements specified herein when installed in vehicle.

3.5.1 Torque and loads. The transmission, operating at 1600 revolutions per minute (rpm), shall transmit an input torque of 217 pounds-foot (lbs-ft) in all gear ratios (see table I) to the output shaft with the appropriate torque multiplication. The output shaft shall transmit a maximum of 1553 lb-ft to the vehicle power train.

3.5.2 Environmental conditions.

3.5.2.1 Low ambient temperature. With transmission filled to the specified level using lubricant of the grade specified in MIL-L-2105, the transmission assembly shall maintain operation when exposed to an ambient temperature of minus 65 degrees Fahrenheit (°F).

3.5.2.2 High ambient temperature. At an ambient temperature of 125°F the transmission shall operate continuously under normal load without the lubricant temperature exceeding 275°F. Transmission shall operate satisfactorily with lubricant temperature not exceeding 300°F under severe operating conditions for 15 minutes maximum. Lubricant temperature shall be measured at the transmission sump.

3.5.2.3 Salt spray. The transmission assembly shall maintain operation during, and after 48 hours of exposure to a 5 percent salt fog atmosphere as specified in method 509 of MIL-STD-810.

3.5.2.4 Sand and dust. The transmission assembly, as installed in the vehicle, shall maintain operation during, and after exposure to sand and dust conditions specified in method 510 of MIL-STD-810.

3.5.3 Steam and water jet cleaning. The transmission, as installed in vehicle, or a simulated mounting with protection to the input and output shafts, shall withstand cleaning with high pressure steam and cleaning compound conforming to P-C-437 and water jet, without deterioration of seals, water leakage past seals, gaskets, or other defects.

3.5.4 Slopes. The transmission shall operate in all gears in any position normal to the as-installed operating attitude limits of 60 percent longitudinal grades, and 30 percent side slopes, without excessive lubricant leakage.

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3.6 Data plates. Unless otherwise specified (see 6.2), data plates shall conform to MIL-P-514.

3.7 Finish. The transmission shall be treated in accordance with type III of TT-C-490 primed in accordance with TT-P-636, and painted olive drab TT-E-529.

3.8 Workmanship. The transmission assembly shall be constructed and finished in a workmanlike manner. Attention shall be paid to neatness and thoroughness of marking parts, plating, painting, machine- screw assemblage, welding and brazing, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Interchangeability of parts. Transmission parts and subassemblies shall be evaluated to determine conformance to 3.3.1.

4.1.2 Materials inspection. Materials inspection shall consist of certification, supported by verifying data, that materials and parts specified in 3.2 used in fabricating transmission assemblies, are in accordance with applicable referenced specifications or requirements.

4.2 Classification of inspections. Classification of inspections shall be as follows:

a. First article (intial production) inspection (see 4.3).
b. Quality conformance inspection (see 4.5).

4.3 First article (initial production) inspection. To determine conformance to Section 3 (inclusive), two transmissions of the first ten produced under the contract selected at random by the Government, shall be subjected to initial production inspection. Both samples of the lot shall be subjected to break-in run (see 4.4) and quality conformance (see 4.5.2). If submitted transmissions meet all requirements of the contract and this specification, no modifications shall be applied to subsequent transmissions without prior approval by the Government.

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4.3.1 Sample I. Upon completion of testing in accordance with 4.3, sample I shall be shipped at contractor's expense, to an installation specified by the Government. The transmission shall be installed in an M561 truck or M792 ambulance. Unless otherwise specified, the vehicle with a rated payload, shall be operated for 20,000 miles and shall be capable of operating as follows:

a. 30 percent on paved roads either concrete or asphalt or any combination of the two.
b. 36 percent on secondary dirt or gravel roads.
c. 30 percent on cross-country terrain.
d. 4 percent on Belgian block.

4.3.2 Reliability verification. To determine conformance to 3.1.1, reliability shall be verified during 20,000 miles of operation of sample I.

4.3.3 Sample II. Upon completion of testing in accordance with 4.3, sample II shall be re-tested as specified in 4.5.3.2, except that the test shall be extended to 10 cycles in each gear. Cycles shall be alternated between 1800 and 2600 rpm. Sample II shall then be tested as specified in table II.

TABLE II. First article testing.

Description	Requirement	Test
Torque and load test	3.5.1	4.6.1
Low ambient temperature test	3.5.2.1	4.6.2
High ambient temperature and lubricant temperature test	3.5.2.2	4.6.3
Salt spray test	3.5.2.3	4.6.4
Sand and dust test	3.5.2.4	4.6.5
Steam and water jet cleaning test	3.5.3	4.6.6
Slope test	3.5.4	4.6.7

4.3.4 Failure. Failure to pass any examination or test, or defects in excess of acceptable quality levels (AQL's) specified, shall be cause for refusal to grant first article approval.

4.4 Break-in run. Each transmission shall be filled to proper level with lubricant conforming to MIL-L-2105, and shall be installed on a device which provides power to the input (clutch) shaft and a simulated load resistance to the output shaft. Each transmission shall be subjected to a run on this device in each gear of sufficient duration to demonstrate basic, safe functioning of the transmission. Input shaft speed for the break-in run shall be 1,000 rpm (nominal) and input torque 50 lb-ft $\pm$ 10 lb-ft. On completion of break-in run, the lubricating oil shall be drained and examined for metal chips or other foreign matter.

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4.4.1 Malfunction. Transmissions showing evidence of mechanical or other malfunction(s) during break-in run shall not be subjected to all testing. Lubricant leakage, abnormal noise, or metal particles found in the drained oil shall also constitute malfunctions.

4.5 Quality conformance inspection.

4.5.1 Lot formation. A lot shall consist of all transmissions from an identifiable production period, submitted at one time for acceptance.

4.5.2 Quality conformance examination.

4.5.2.1 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with inspection level II of MIL-STD-105.

4.5.2.2 Acceptable quality level. Each transmission selected in accordance with 4.5.2.1 shall be examined for conformance to the following acceptable quality levels (AQL), on the basis of percent defective.

Classification	AQL
Major	2.0
Minor	6.5

4.5.2.3 Classification of defects. For examination purposes, defects shall be classified as specified in table III. Acceptable quality levels shall be as specified in table III.

TABLE III. Classification of defects.

Category	Defect	Method of inspection
Major	Major AQL = 2.0 percent	
101	Dimensions affecting interchangeability, not within tolerance (see 3.3 and 3.3.1).	Gage or visual
102	Oil leakage during or after any test(s) in excess of a seep (see 3.3.3, 3.5.4 and 6.4).	Visual
103	Not following good workmanship practice(s), improper installation or adjustment of components (see 3.8).	Visual
Minor	Minor AQL = 6.5 percent	
201	Dimensions not affecting interchangeability, not within tolerance (see 3.3 and 3.3.1).	Gage or visual
202	Improper or illegible marking (see 3.6).	Visual
203	Improper application or color of paint (see 3.7).	Visual

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4.5.2.4 Unclassified defects. All defects having no bearing on function, safety, interchangeability, or life; but which are considered departures from good workmanship, will be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots, or 10 lots within a 30 day period, will be added to the minor classification of defects with no increase in AQL's (see 6.5).

4.5.3 Quality conformance test.

4.5.3.1 Acceptance tests.

4.5.3.1.1 Sampling for acceptance testing. Each transmission of the lot shall be tested.

4.5.3.1.2 Acceptance test. Each transmission shall be filled to the proper level with specified lubricant (see 3.2.1), installed on a suitable test fixture, and tested in accordance with table IV. Input torque for the test shall be not less than 100 lb-ft in all gears.

TABLE IV. Acceptance test.

Cycle	Gear	Minutes at 1800 rpm	Minutes at 2600 rpm
1	First	2	2
2	Second	3	3
3	Third	3	3
4	Fourth	1	1
5	Reverse	1	1

During the test, the transmission shall be examined for unusual noises, ease of gearshifting, and lubricant leakage.

4.5.3.1.3 Acceptance test failure. Failure of any transmission to pass any acceptance examination or test, shall be cause for the Government to refuse acceptance of subsequent lots until the defect(s) revealed by the examination or test have been corrected.

4.5.3.2 Control test. A control test shall be conducted by the contractor to determine that quality of the transmission is maintained.

4.5.3.2.1 Sampling for control test. Transmissions shall be selected at random by the Government from transmissions that have passed acceptance inspection at the rate of two per month when production is 200 units or less per month; and at the rate of four per month, when production is greater than 200 units per month. Transmissions shall be identified as to production period and acceptance test lot.

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4.5.3.2.2 Control test routine. The selected transmission shall be filled to the proper level with the lubricant specified in 3.2.1, installed in a suitable test fixture, and tested in accordance with table V.

TABLE V. Control test.

Gear	Hours	Cycles	Torque at 1800 rpm	Torque at 2600 rpm
1	2	4	210 lb-ft	170 lb-ft
2	3	4	210 lb-ft	170 lb-ft
3	3	4	210 lb-ft	170 lb-ft
4	1	4	210 lb-ft	170 lb-ft
R	1	4	125 lb-ft	50 lb-ft

The first and third cycles shall be run at 1800 rpm, the second and fourth cycles at 2600 rpm. The test shall be run on a continuous basis. Lubricating oil temperature shall be monitored and shall conform to 3.5.2.2. During the test, the transmission shall be examined for unusual noises, lubricant leakage, and ease of gearshifting.

4.5.3.2.3 Control test failure. Failure of a control test sample to pass specified test shall be cause for the Government to refuse acceptance of subsequent lots, until it has been proved to the satisfaction of the Government that the faults revealed by the test have been corrected.

4.5.3.3 Comparison test. To determine conformance to Section 3 (inclusive), the Government may select transmissions at any time during the contract production period, and subject transmissions to applicable tests. Tests will be conducted by the Government at a site selected or approved by the Government. Transmissions selected for proving ground tests will not include any transmission previously control tested.

4.5.3.3.1 Comparison test failure. Failure of any transmission(s) tested to conform to any requirement will cause the Government to discontinue acceptance until corrective action has been implemented.

4.6 Methods of inspection.

4.6.1 Torque and loads test. To determine conformance to 3.5.1, the transmisson shall be filled to proper level with the lubricant specified in 3.2.1, and installed in a suitable test fixture. The transmission shall be run in first gear for 15 minutes under the following simultaneous conditions:

a. Input speed shaft - 1600 rpm.
b. Input torque - 220 lb-ft
c. Output torque - 1553 lb-ft

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4.6.2 Low ambient temperature test. To determine conformance to 3.5.2.1, the transmission shall be filled to the specified level with lubricant of grade specified for sub-zero temperatures, and placed on a suitable test fixture in a temperature controlled cell. The transmission shall be cold-soaked to -65°F. The transmission shall then be tested in an ambient temperature of -65°F ± 5°F. The test routine shall be that specified in 4.5.3.1.2.

4.6.3 High ambient temperature test. To determine conformance to 3.5.2.2, the transmission shall be filled to the specified level with lubricant of grade specified for high ambient temperature, and placed on a suitable test fixture in a temperature controlled cell. The transmission shall be heat-soaked to 125°F. The transmission shall then be tested in the ambient temperature of 125°F ± 5°F. The test routine shall be that specified in 4.5.3.2.2, except test hours shall be reduced 50 percent. Lubricant temperature shall be monitored and shall remain within limits specified.

4.6.4 Salt spray test. To determine conformance to 3.5.2.3, the transmission shall be subjected to test specified in method 509 of MIL-STD-810.

4.6.5 Sand and dust test. To determine conformance to 3.5.2.4, the transmission shall be subjected to test specified in method 510 of MIL-STD-810.

4.6.6 Steam and water jet cleaning test. To determine conformance to 3.5.3, the transmission shall be subjected to cleaning as specified. Steam and water jet cleaning is defined as follows: The jet is applied perpendicular to the surface to be cleaned at a minimum distance of one foot from the surface. Cleaning is accomplished at a minimum rate of one square foot per minute. The jet pressure shall be not less than 100 pounds per square inch (psi) nor more than 110 psi. The seals and gaskets shall be examined for damage and leaks (see 3.3.3). Paint removal shall not be considered a deficiency.

4.6.7 Slope test. To determine conformance to 3.5.4, the transmission shall be mounted in a suitable fixture and tested at simulated grades and slopes specified. After the test, the transmission shall be examined for excessive lubricant leakage.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level of protection shall be in accordance with the applicable packaging standard or packaging data sheet specified by the acquisitioning activity (see 6.2).

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6. NOTES

6.1 Intended use. The transmission assembly will be used to transmit power from the engine to the M561 or M792 vehicle drive train.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If data plates shall conform to other than MIL-P-514.
c. Selection of method and applicable level of preservation and packaging, and level of packing of the referenced classification (see 5.1).

6.3 First article. First article samples shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for examination, tests, and approval of the first article (see 3.1).

6.4 Classification of leakage defects. The following definitions shall be used for the classification of leakage defects:

a. Weep: - Any non-recurring evidence of fluid beyond the seal.
b. Seep: - Any recurring evidence of fluid beyond the seal that does not result in the formation of a droplet.
c. Droplet: - Any recurring evidence of fluid beyond the seal that results in the formation of a non-falling droplet.
d. Drip: - Any recurring evidence of fluid beyond the seal where a droplet forms and falls.

6.5 Deficiency sheet. The Government inspector shall verify that a thorough inspection of each transmission is performed by the contractor for the listed characteristics, and for any departures from good workmanship. The Government inspector shall assure that all defects encountered during inspection are enumerated on Deficiency Sheet for the transmission. Defects noted on a Deficiency Sheet shall contain sufficient description to enable the Government inspector and the contractor's representative to classify the defects in accordance with the classification of defects (see table II) and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring defects (see 4.5.2.4).

6.5.1 Recurring major defects. A major defect is recurring when the same defect occurs more than once in the same sample, or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or equivalent) reflect such a condition. Recurring major defects shall be cause for the entire lot or lots to be inspected for the recurring defects.

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6.5.2 Recurring minor defects. A minor defect is recurring if it occurs more than twice in the same sample, or when the defect occurs in four successive samples. Recurring minor defects shall be cause for the entire lot or lots to be inspected for the recurring defects.

6.6 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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